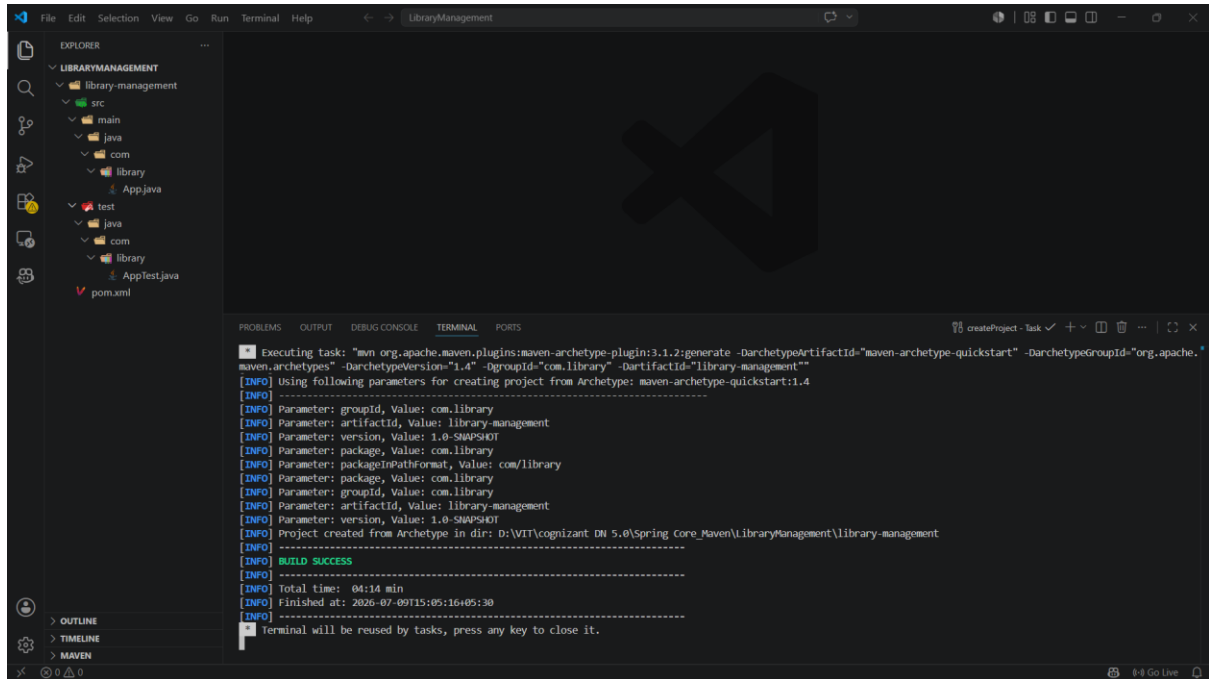


Spring Core Maven Exercise 1 | Cognizant DN 5.0

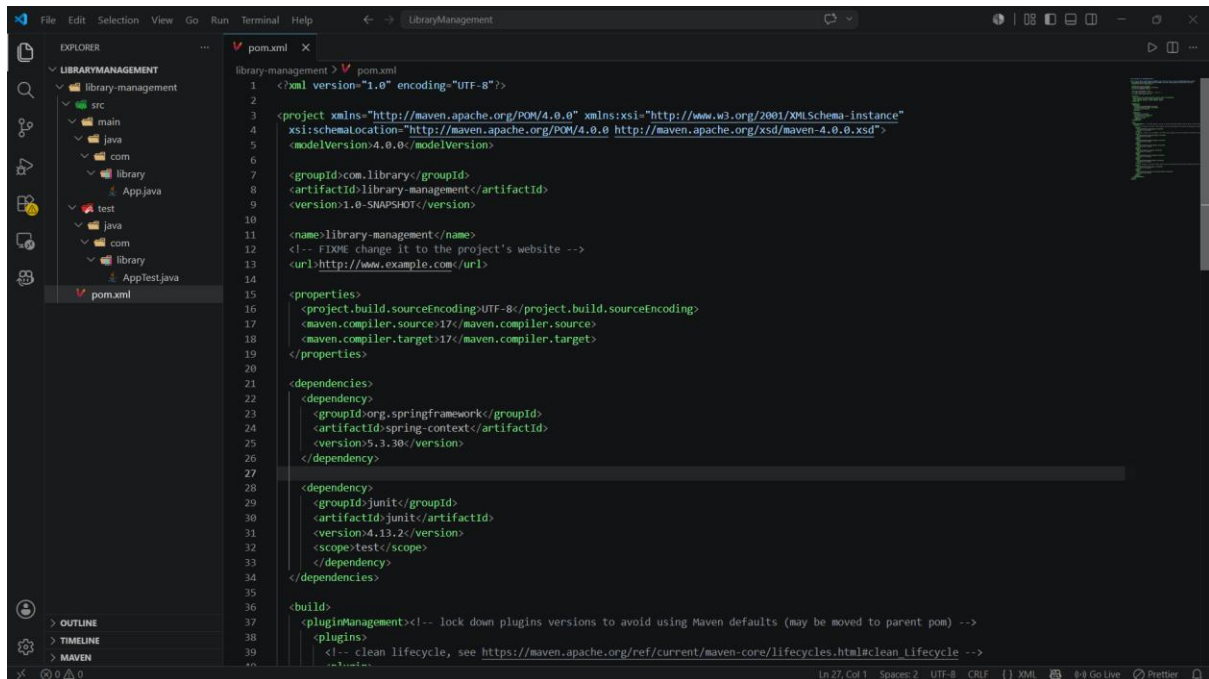
SANSKAR KUMAWAT

EXERCISE 1 - Configuring a Basic Spring Application

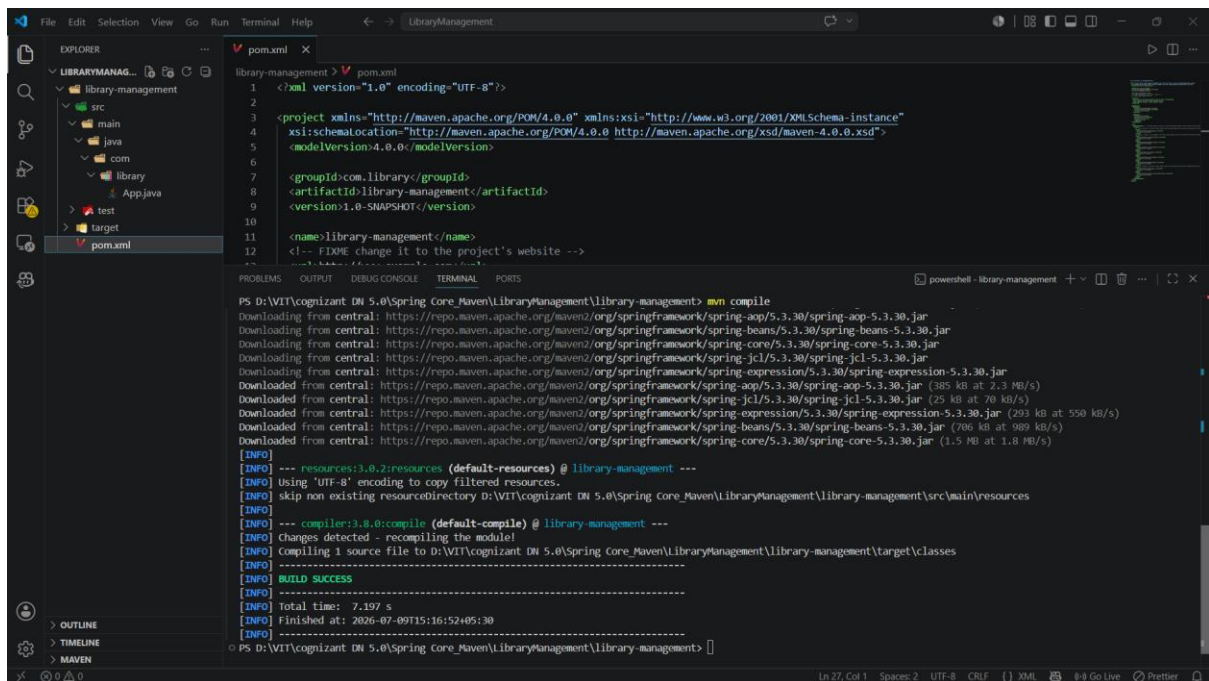
Create new Maven project LibraryMagement (library-management) –



Update pom.xml –



Mvn compile -

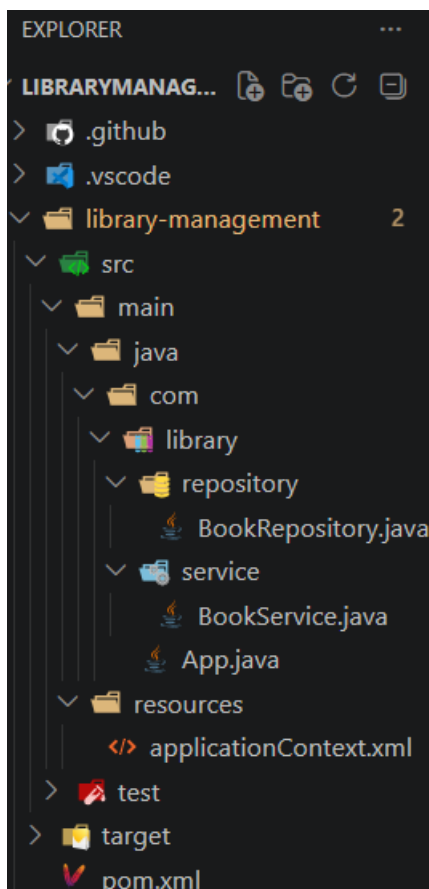


The screenshot shows the VS Code interface with the Maven compile command executed in the terminal. The Explorer view on the left shows the project structure: library-management, src, main, java, com, library, repository, service, resources, test, target, and pom.xml. The pom.xml file is open in the editor, showing the Maven configuration. The terminal output shows the download of various Spring Framework dependencies and the successful compilation of the project.

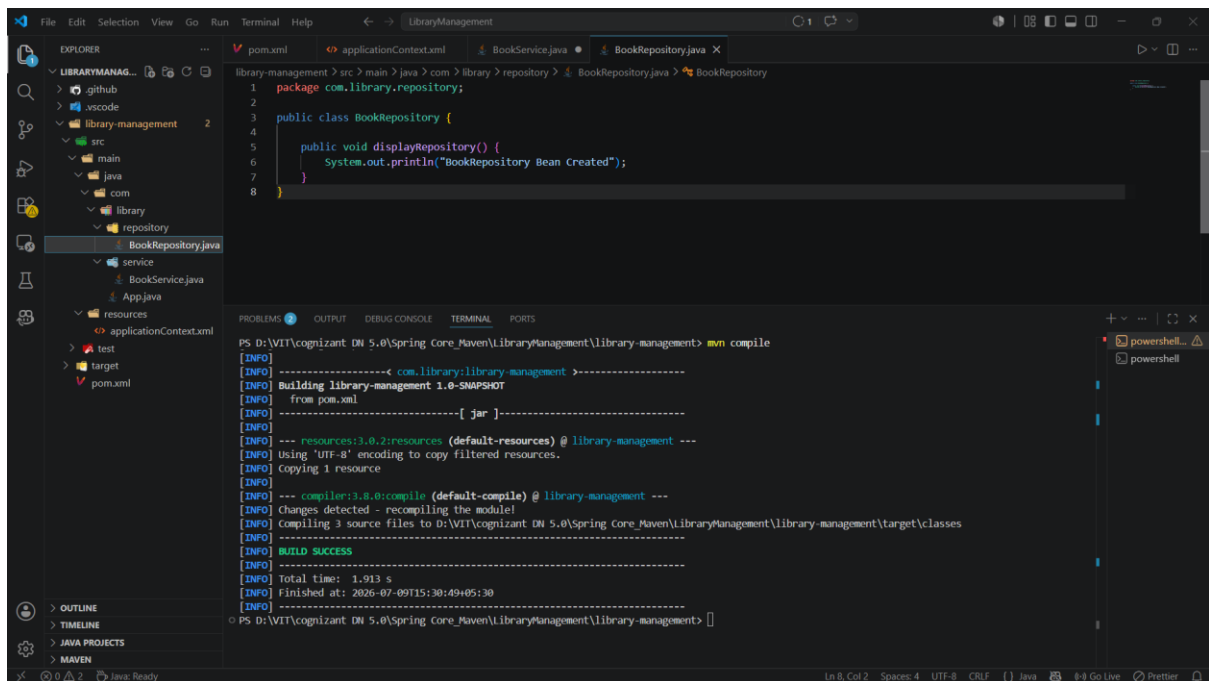
```
library-management > pom.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
5   <modelVersion>4.0.0</modelVersion>
6
7   <groupId>com.library</groupId>
8   <artifactId>library-management</artifactId>
9   <version>1.0-SNAPSHOT</version>
10
11   <name>library-management</name>
12
13   <!-- FIXME change it to the project's website -->
14   <url>http://example.com</url>
15
16   <properties>
17     <java.version>1.8</java.version>
18   </properties>
19
20   <dependencies>
21     <dependency>
22       <groupId>org.springframework</groupId>
23       <artifactId>spring-core</artifactId>
24       <version>5.3.30</version>
25     </dependency>
26     <dependency>
27       <groupId>org.springframework</groupId>
28       <artifactId>spring-beans</artifactId>
29       <version>5.3.30</version>
30     </dependency>
31     <dependency>
32       <groupId>org.springframework</groupId>
33       <artifactId>spring-jcl</artifactId>
34       <version>5.3.30</version>
35     </dependency>
36     <dependency>
37       <groupId>org.springframework</groupId>
38       <artifactId>spring-expression</artifactId>
39       <version>5.3.30</version>
40     </dependency>
41     <dependency>
42       <groupId>org.springframework</groupId>
43       <artifactId>spring-aop</artifactId>
44       <version>5.3.30</version>
45     </dependency>
46     <dependency>
47       <groupId>org.springframework</groupId>
48       <artifactId>spring-jcl</artifactId>
49       <version>5.3.30</version>
50     </dependency>
51     <dependency>
52       <groupId>org.springframework</groupId>
53       <artifactId>spring-expression</artifactId>
54       <version>5.3.30</version>
55     </dependency>
56     <dependency>
57       <groupId>org.springframework</groupId>
58       <artifactId>spring-beans</artifactId>
59       <version>5.3.30</version>
60     </dependency>
61     <dependency>
62       <groupId>org.springframework</groupId>
63       <artifactId>spring-core</artifactId>
64       <version>5.3.30</version>
65     </dependency>
66   </dependencies>
67
68   <build>
69     <plugins>
70       <plugin>
71         <groupId>org.springframework</groupId>
72         <artifactId>spring-maven-plugin</artifactId>
73         <version>5.3.30</version>
74       </plugin>
75     </plugins>
76   </build>
77
78   <resources>
79     <resource>
80       <directory>src/main/resources</directory>
81       <filtering>true</filtering>
82     </resource>
83   </resources>
84
85   <testResources>
86     <resource>
87       <directory>src/test/resources</directory>
88       <filtering>true</filtering>
89     </resource>
90   </testResources>
91
92   <profiles>
93     <profile>
94       <id>dev</id>
95       <activation>
96         <activeByDefault>true</activeByDefault>
97       </activation>
98       <properties>
99         <spring.profiles.active>dev</spring.profiles.active>
100      </properties>
101    </profile>
102  </profiles>
103</project>
```

```
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn compile
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-aop/5.3.30/spring-aop-5.3.30.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-beans/5.3.30/spring-beans-5.3.30.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-core/5.3.30/spring-core-5.3.30.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-jcl/5.3.30/spring-jcl-5.3.30.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-expression/5.3.30/spring-expression-5.3.30.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-aop/5.3.30/spring-aop-5.3.30.jar (385 KB at 2.3 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-jcl/5.3.30/spring-jcl-5.3.30.jar (25 KB at 70 KB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-expression/5.3.30/spring-expression-5.3.30.jar (283 KB at 550 KB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-beans/5.3.30/spring-beans-5.3.30.jar (706 KB at 989 KB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-core/5.3.30/spring-core-5.3.30.jar (1.5 MB at 1.8 MB/s)
[INFO] --- resources:3.0.2:resources (default-resources) @ library-management ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management\src\main\resources
[INFO] --- compiler:3.8.0:compile (default-compile) @ library-management ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management\target\classes
[INFO] BUILD SUCCESS
[INFO] Total time: 7.197 s
[INFO] Finished at: 2026-07-09T15:16:52+05:30
[INFO]
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management>
```

File structure now –



Compiling BookRepository –



```
library-management > src > main > java > com > library > repository > BookRepository.java > BookRepository
```

```
1 package com.library.repository;
```

```
2
```

```
3 public class BookRepository {
```

```
4
```

```
5     public void displayRepository() {
```

```
6         System.out.println("BookRepository Bean Created");
```

```
7     }
```

```
8 }
```

```
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn compile
```

```
[INFO]
[INFO] -----< com.library:library-management >-----
```

```
[INFO] Building library-management 1.0-SNAPSHOT
```

```
[INFO] from pom.xml
```

```
[INFO] -----[ jar ]-----
```

```
[INFO]
```

```
[INFO] --- resources:3.0.2:resources (default-resources) @ library-management ---
```

```
[INFO] Using 'UTF-8' encoding to copy filtered resources.
```

```
[INFO] Copying 1 resource
```

```
[INFO]
```

```
[INFO] --- compiler:3.8.0:compile (default-compile) @ library-management ---
```

```
[INFO] Changes detected - recompiling the module!
```

```
[INFO] Compiling 3 source files to D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management\target\classes
```

```
[INFO]
```

```
[INFO] BUILD SUCCESS
```

```
[INFO]
```

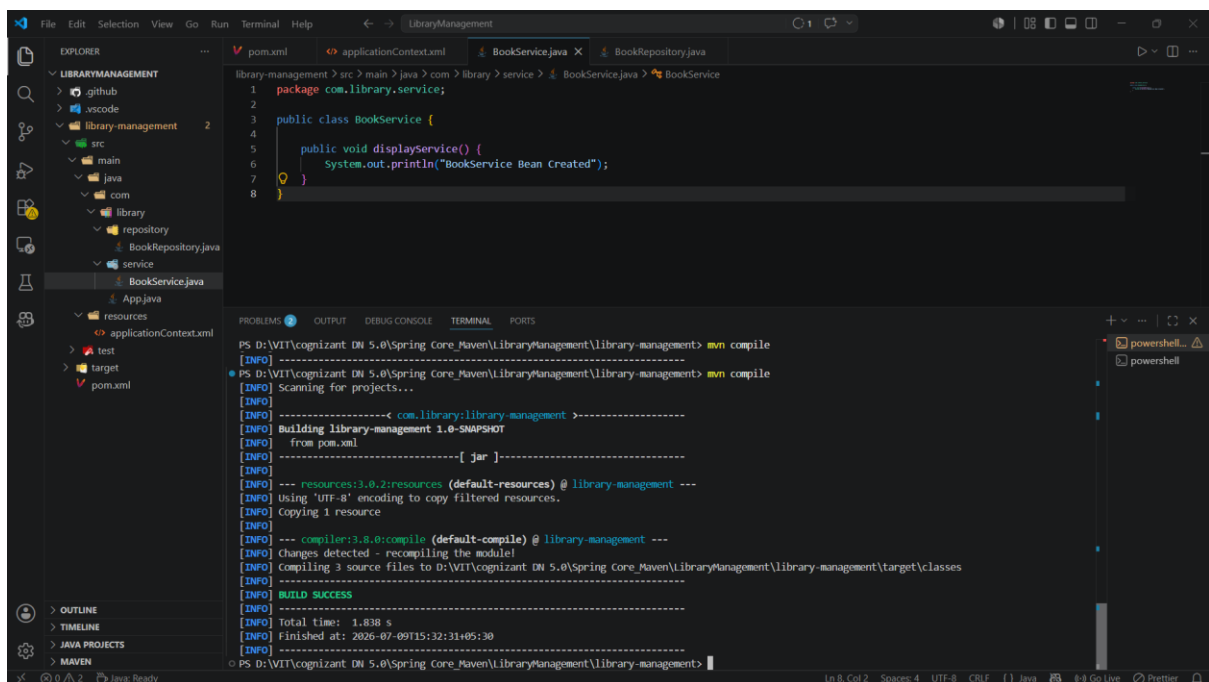
```
[INFO] Total time: 1.913 s
```

```
[INFO] Finished at: 2026-07-09T15:30:49+05:30
```

```
[INFO]
```

```
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management>
```

Compiling BookServices –



```
library-management > src > main > java > com > library > service > BookService.java > BookService
```

```
1 package com.library.service;
```

```
2
```

```
3 public class BookService {
```

```
4
```

```
5     public void displayservice() {
```

```
6         System.out.println("BookService Bean Created");
```

```
7     }
```

```
8 }
```

```
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn compile
```

```
[INFO]
```

```
[INFO] -----< com.library:library-management >-----
```

```
[INFO] Building library-management 1.0-SNAPSHOT
```

```
[INFO] from pom.xml
```

```
[INFO] -----[ jar ]-----
```

```
[INFO]
```

```
[INFO] --- resources:3.0.2:resources (default-resources) @ library-management ---
```

```
[INFO] Using 'UTF-8' encoding to copy filtered resources.
```

```
[INFO] Copying 1 resource
```

```
[INFO]
```

```
[INFO] --- compiler:3.8.0:compile (default-compile) @ library-management ---
```

```
[INFO] Changes detected - recompiling the module!
```

```
[INFO] Compiling 3 source files to D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management\target\classes
```

```
[INFO]
```

```
[INFO] BUILD SUCCESS
```

```
[INFO]
```

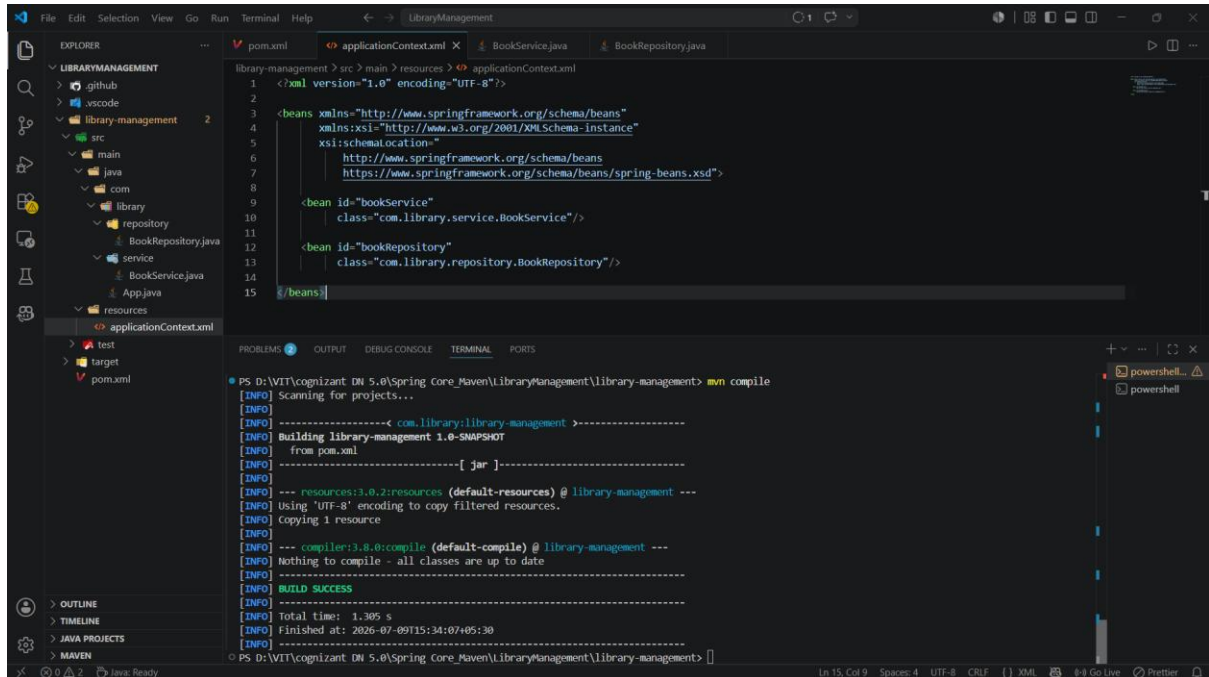
```
[INFO] Total time: 1.838 s
```

```
[INFO] Finished at: 2026-07-09T15:32:31+05:30
```

```
[INFO]
```

```
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management>
```

Configuring applicationContext.xml -



The screenshot shows the VS Code interface with the `applicationContext.xml` file open in the editor. The XML configuration defines two beans: `bookService` and `bookRepository`. The terminal window shows the output of the `mvn compile` command, which successfully builds the project.

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
         http://www.springframework.org/schema/beans
         https://www.springframework.org/schema/beans/spring-beans.xsd">

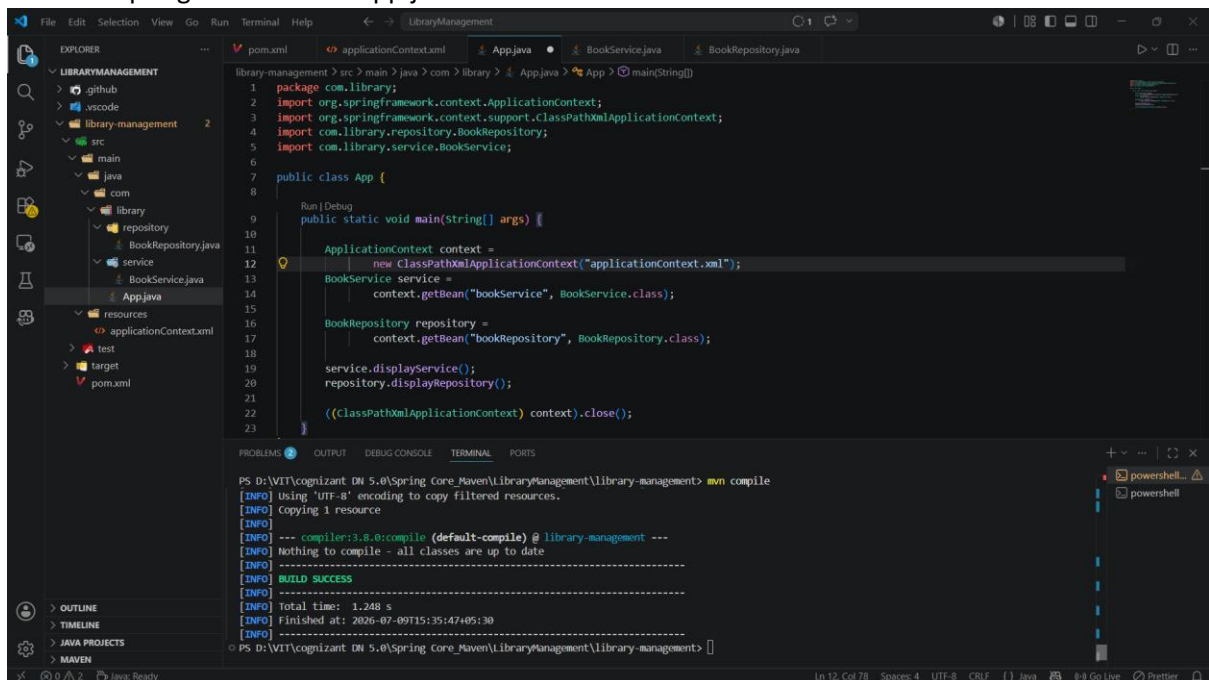
    <bean id="bookService"
          class="com.library.service.BookService"/>

    <bean id="bookRepository"
          class="com.library.repository.BookRepository"/>

</beans>
```

```
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ----- com.library:library-management -----
[INFO] Building library-management 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ library-management ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ library-management ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] ----- BUILD SUCCESS -----
[INFO]
[INFO] Total time: 1.305 s
[INFO] Finished at: 2026-07-09T15:34:07+05:30
[INFO]
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management>
```

Load the spring Container in App.java -



The screenshot shows the VS Code interface with the `App.java` file open in the editor. The code imports the necessary Spring classes and creates an `ApplicationContext` to load the beans defined in `applicationContext.xml`. The terminal window shows the output of the `mvn compile` command, which successfully builds the project.

```
package com.library;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.repository.BookRepository;
import com.library.service.BookService;

public class App {

    Run | Debug
    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            context.getBean("bookService", BookService.class);

        BookRepository repository =
            context.getBean("bookRepository", BookRepository.class);

        service.displayService();
        repository.displayRepository();

        ((ClassPathXmlApplicationContext) context).close();
    }
}
```

```
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn compile
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ library-management ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] ----- BUILD SUCCESS -----
[INFO]
[INFO] Total time: 1.248 s
[INFO] Finished at: 2026-07-09T15:35:47+05:30
[INFO]
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management>
```

```
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn exec:java
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-exec/1.3/commons-exec-1.3.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-exec/1.3/commons-exec-1.3.jar (54 kB at 182 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-component-annotations/2.1.1/plexus-component-annotations-2.1.1.jar (4.1 kB at 46 kB/s)
BookService Bean Created
BookRepository Bean Created
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.848 s
[INFO] Finished at: 2026-07-09T15:40:25+05:30
[INFO]
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management>
```

BookService Bean created and BookRepository Bean Created...successfully.

